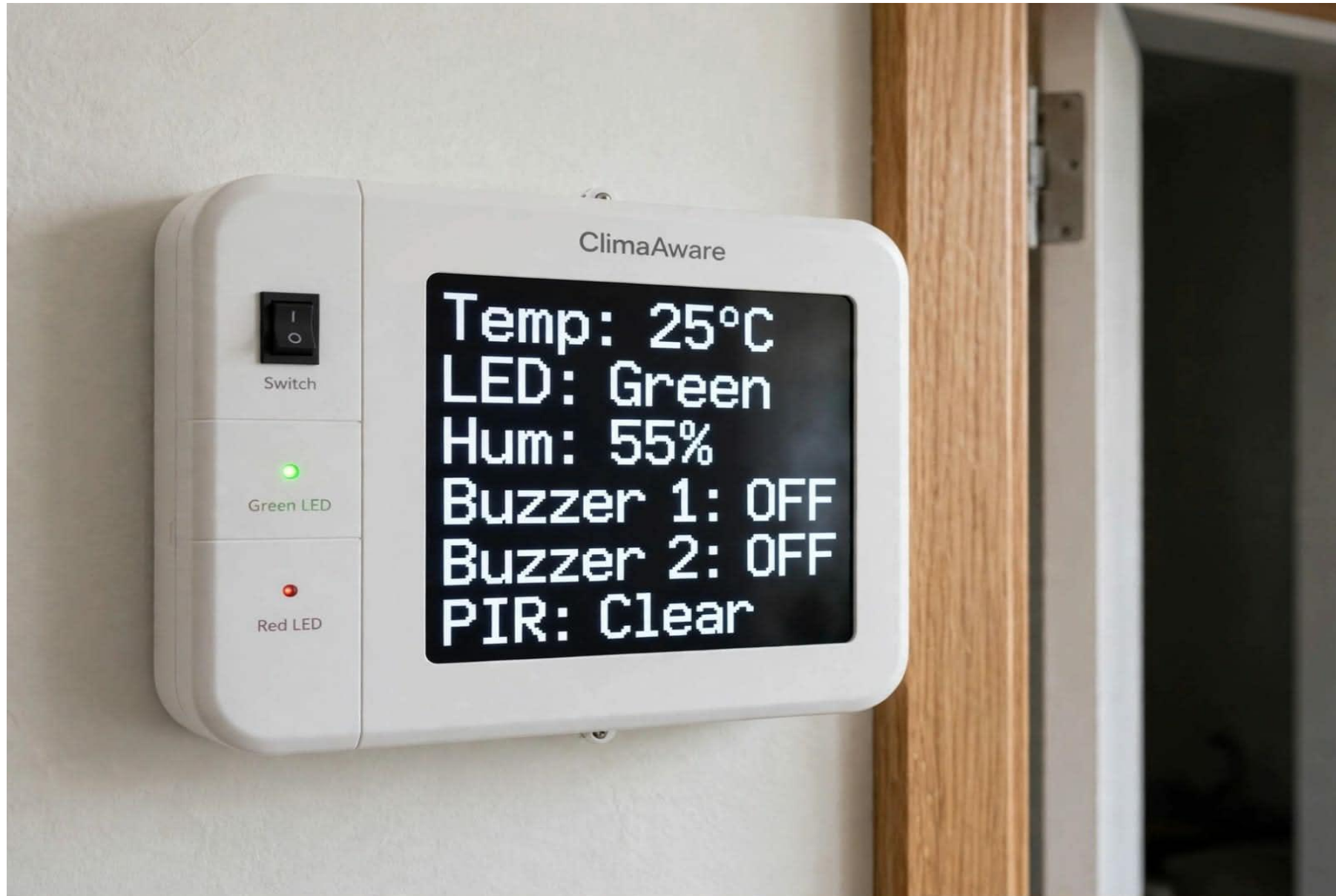


Mechanical Design



My ClimaAware device is housed in a custom enclosure designed to be mounted at a child's exit door.

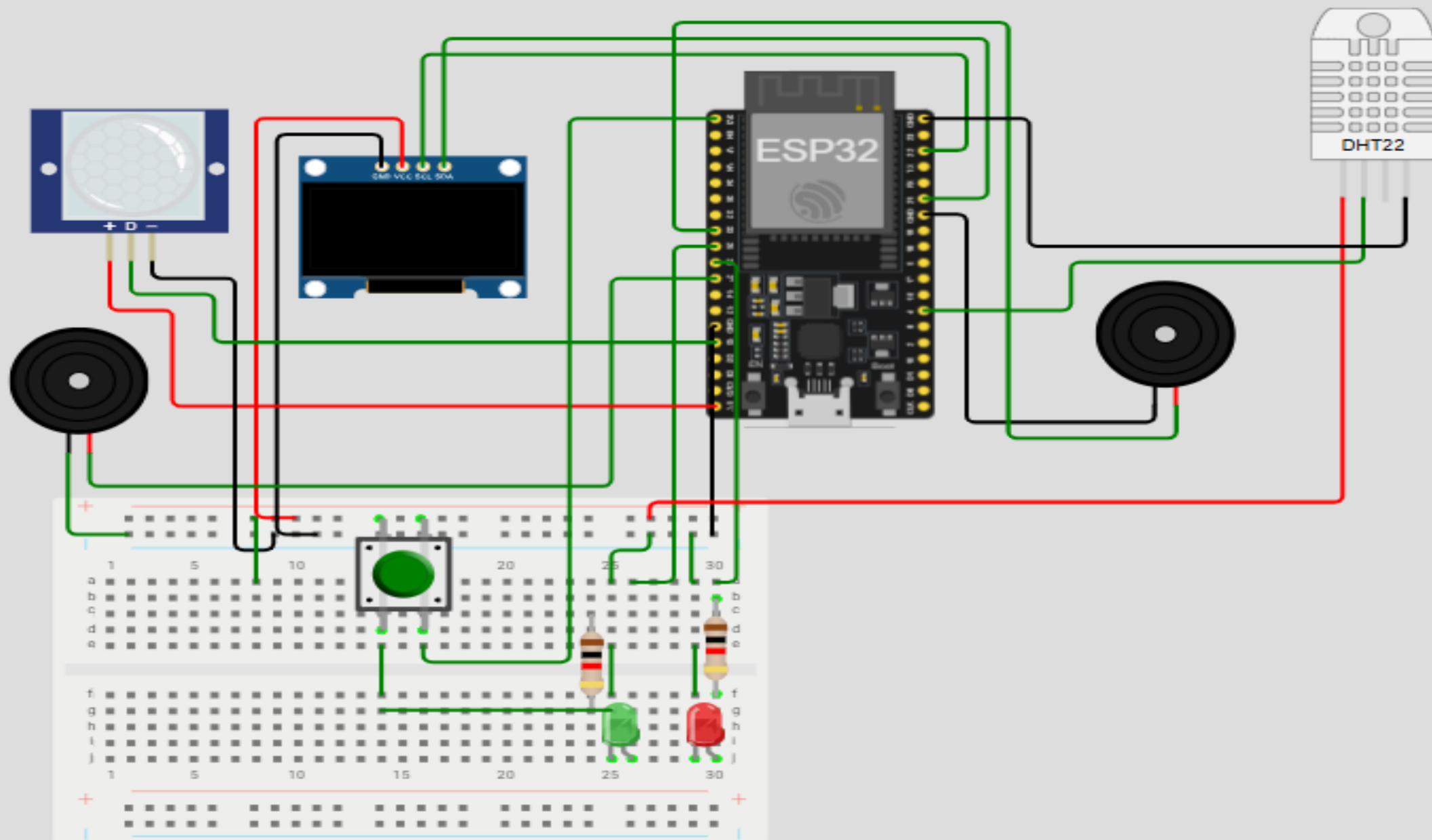
Build & Fabrication:

- The enclosure is 3D printed using PLA plastic. It is designed with ventilation slots on the sides for accurate DHT22 temperature/humidity reading, a front window for the OLED display and Green/Red status LEDs, and a bottom opening for the PIR motion sensor to detect children. The box is split into top and bottom halves fastened with M3 screws for easy maintenance.

How components are connected

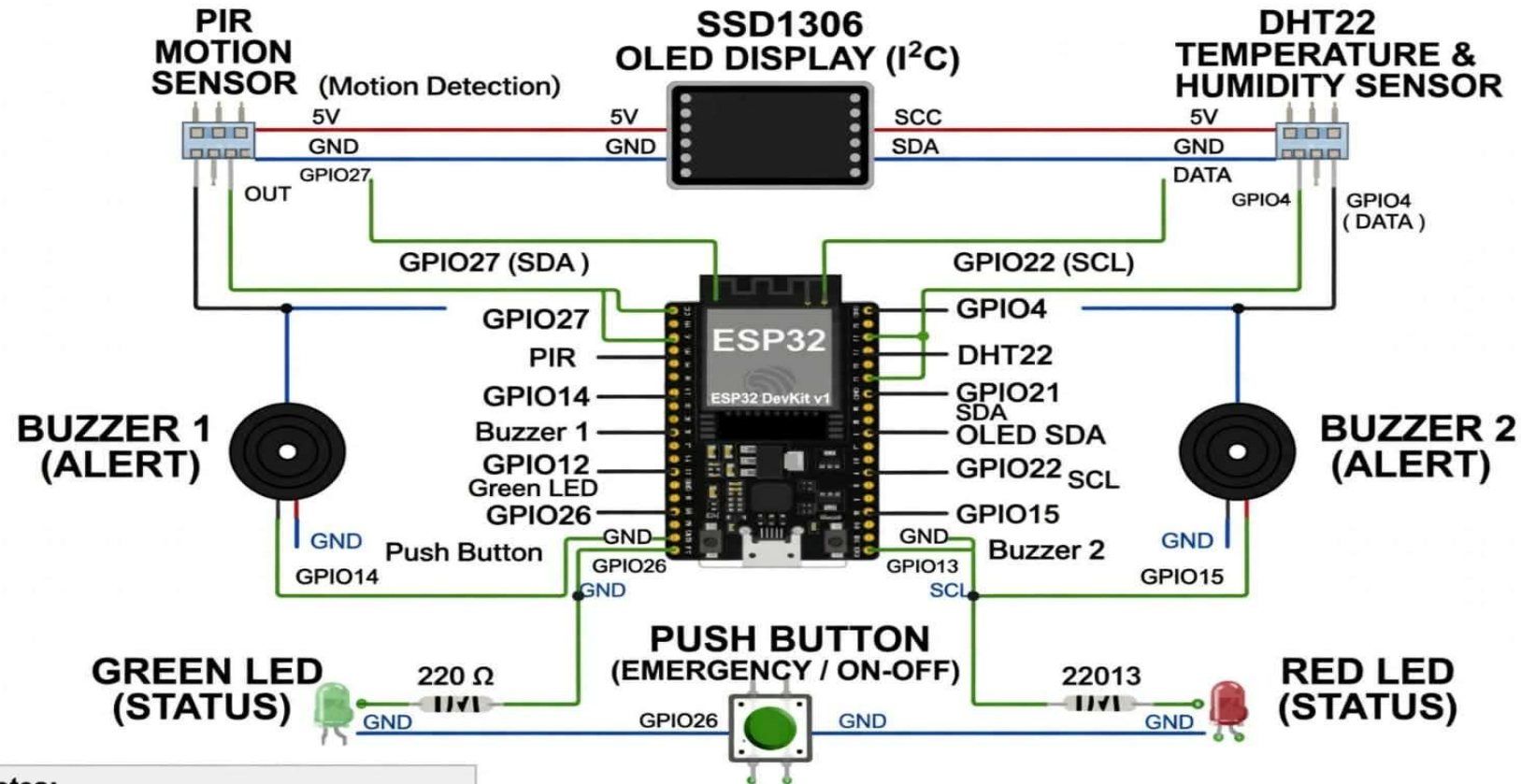
The ESP32 is powered through a rocker switch that acts as the ON/OFF button connecting USB 5V to the ESP32 VIN and the 5V rail for all components with a common GND. The DHT22 DATA is connected to GPIO4, the OLED SSD1306 SDA to GPIO21 and SCL to GPIO22, the PIR sensor OUT to GPIO27, the Green LED with a 220Ω resistor to GPIO12, the Red LED with 220Ω to GPIO13, Buzzer 1 to GPIO14, Buzzer 2 to GPIO15, and the Emergency Stop push button connects GPIO26 to GND when pressed.

(click to edit)



Schematic Design

ClimaAware - Schematic Diagram



Notes:

- All VCC pins connected to 5V rail, all GND pins connected to GND rail
- 220 Ω resistors used to limit current for LEDs
- Push Button is active LOW; ESP32 GPIO26 uses internal pull-up

Power: ESP32 powered via USB 5V